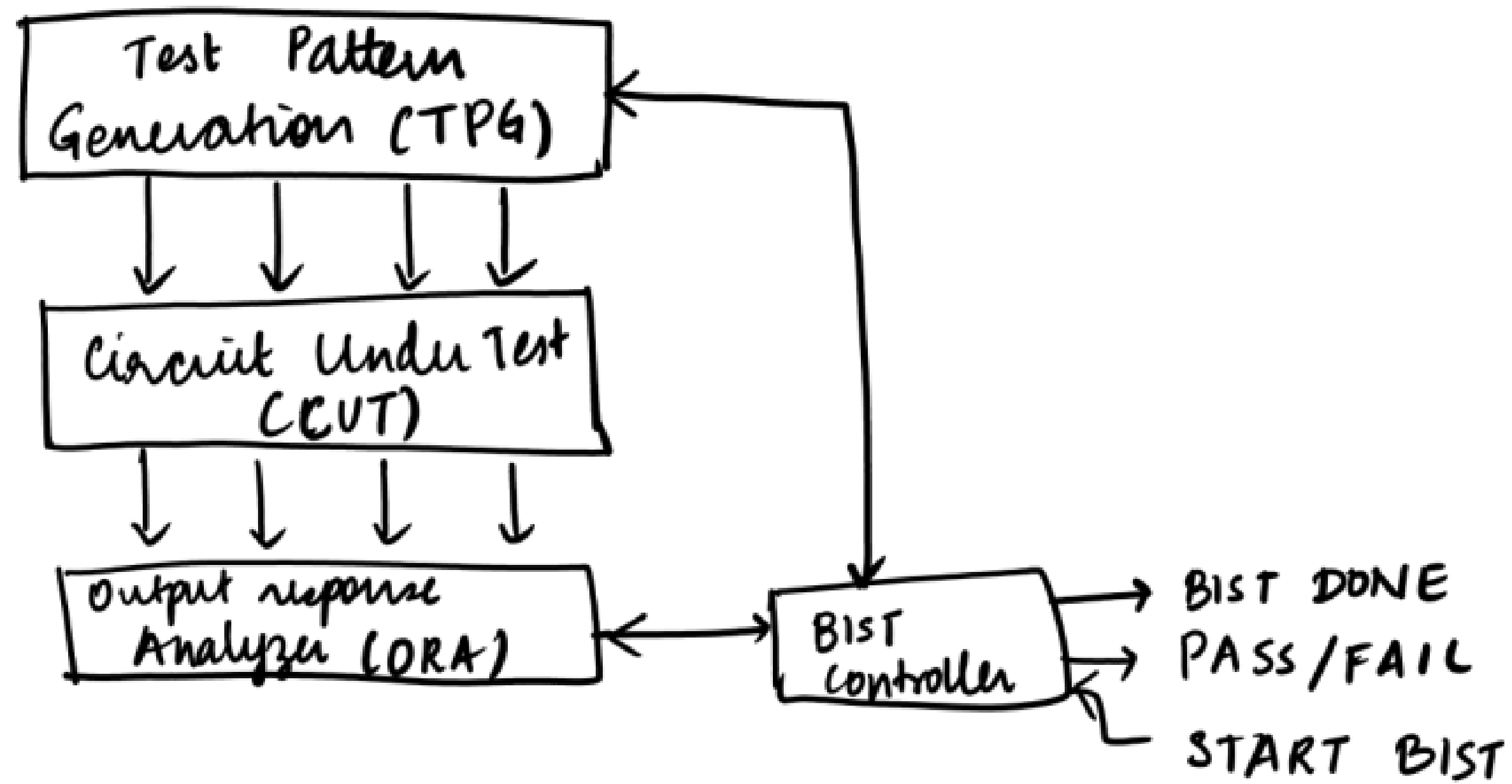


LBIST

Disclaimer : I have just shared my views on LBIST. I have not practically worked on LBIST. If you need to learn LBIST for clearing interviews, please contact institute management to provide trainings on LBIST.

LOGIC BIST :



When we give a start BIST signal, the BIST controller will control the TPG to generate the test patterns to the CUT. The ORA will observe the o/p of the CUT and compare them with expected good outputs. The result will be returned to the BIST controller. Once the BIST is done, the BIST DONE signal will be asserted and the result of the test is a pass/fail signal.

LBIST is mainly targeted for in-system self-test (like automotive safety). It allows the chip to test itself without an external tester.

Example : Automobile safety standards demand that chips be able to test themselves periodically while in operation.

For example:

When you start your car, the chip must run a power-on self-test to ensure no latent fault exists before enabling safety systems (airbag etc.).

During operation, it may need to run online periodic self-tests without external equipment (ATE).

LBIST provides this because it works without a tester/ATE.

This ensures the system knows it's "healthy" *before* it's allowed to control safety mechanisms.